IHP-Open-PDK Review: Present Status and Future Directions

Krzysztof Herman

Herramientas Open-Source para desarrollo de circuitos integrados

Projects: BMFTR -> FMD-QNC (16ME0831)
VDE/VDI -> IHP Open130-G2 (16ME0852)

Leibniz Institute for high performance microelectronics

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Introduction to the IHP

Our motivation and goals in the open source silicon domain

Description of the open source soultions developed and supported by IHP

Understanding the power of the community

Planification of Open MPW runs

Leassons learned and future activities

IHP in a nutshell

- IHP is the European research and innovation centre for silicon-based systems, ultrahigh-frequency circuits and technologies,
- Unique selling point of a 200mm pilot line for state-of-the-art BiCMOS technologies, operated under industry-like conditions, 24/7, for the provision of prototypes and low-volume production runs.
- Qualified technological platform with direct access for science and industry
- Vertical structure from material research to system architecture
- 350+ employees, 40+ nationalities



Vision

"We create foundations and prototype applications based on future silicon-based technologies and systems for a digitalized and networked world as well as for the sustainable preservation of our natural living conditions."

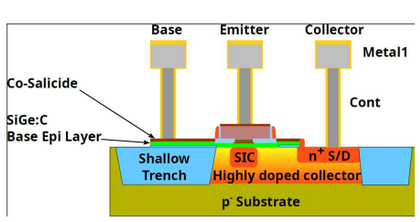
130nm SiGe BiCMOS Technologies for RF Applications

	SG13S	SG13G2	SG13G3Cu
HBT f_t / f_{max}	250 / 340 GHz	350 / 450 GHz	470 / 650 GHz
$W_{Emitter}$	170 nm	130 nm	110 nm
HBT BV_{CE0}	1.7 V	1.6 V	1.5 V
CMOS node	130 nm		
Active devices	Schottky diodes, Antenna diodes, PN diodes, ESD		
Varactors	NMOS Varactor		
ors	Poly-Si, Thin Film		Poly-Si
MIM Caps	1.5 fF / µm² (Al) 2.1 fF / µm² (Cu)	1.5 fF / µm² (Al) 2.1 fF / µm² (Cu)	2.1 fF / µm²
Metallization	7 Layers AL incl. 2 & 3 µm layers or *Cu: 4 + 2 (3µm) Al: 2 (3µm)	7 Layers AL incl. 2 & 3 µm layers or *Cu: 4 + 2 (3µm) Al: 2 (3µm	*Cu: 4 + 2 (3µm) Al: 2 (3µm

*Cu BEOL from X FAB

- Target are high-end technology developments, low volume market introduction, technology transfer for potential mass production in commercial fabs
- SG13S & SG13G2 are qualified and ready for Low Volume of high end products
- SG13G3Cu is early access - qualification scheduled 2025

- SG13G2 technology was selected for the development of an open source PDK



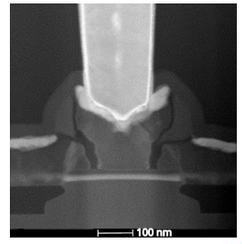


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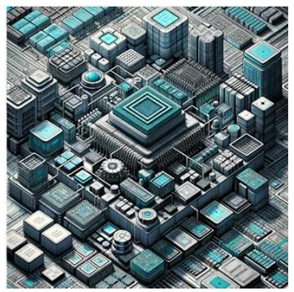
Planification of Open MPW runs

Leassons learned and future activities

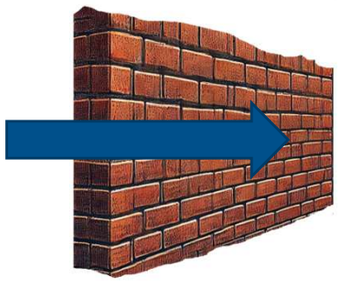
Traditional ASIC development (academic perspective)

EDA provider

NDA walls

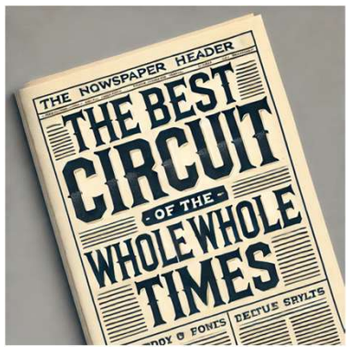


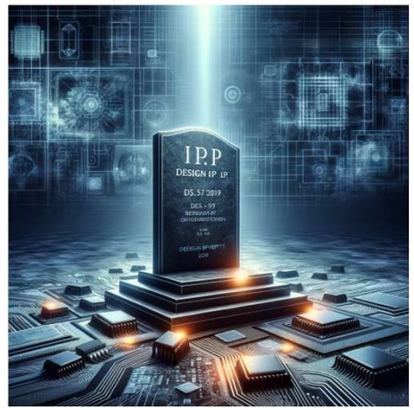
Existig IP blocks





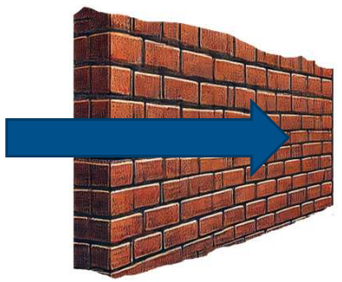
Designer

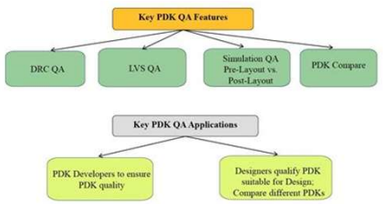






Foundry

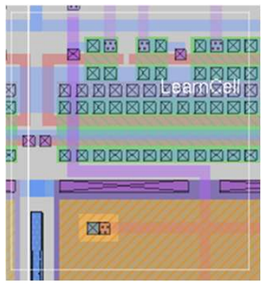




PDK

Flexibility is wanted

Open source ASIC development as an alternative

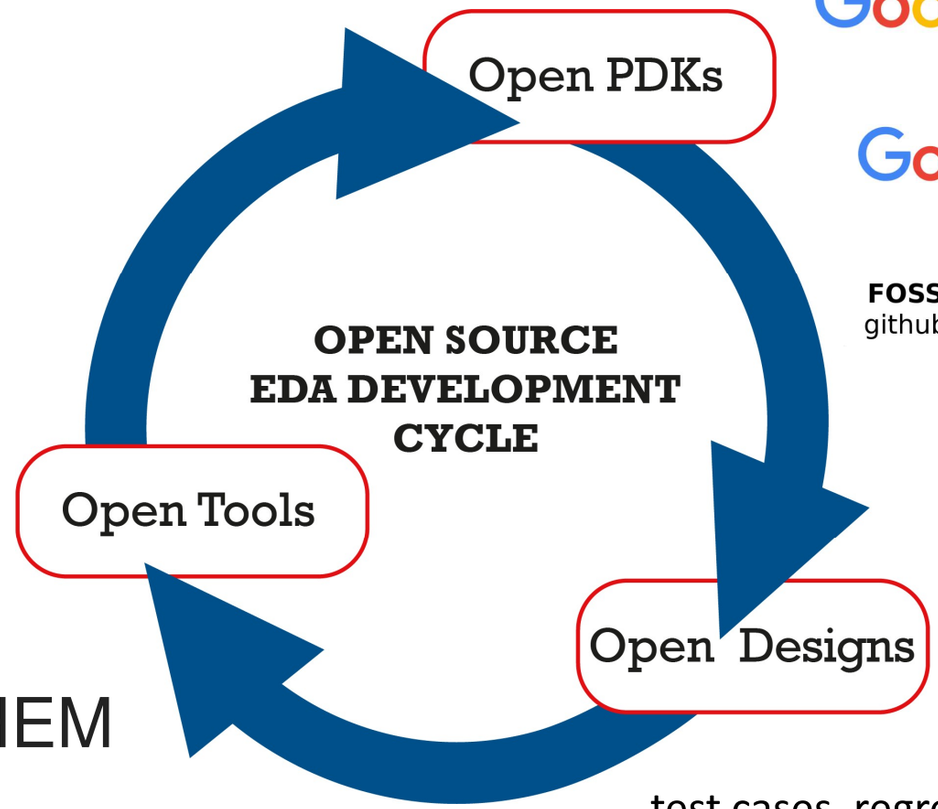


Qucs

NGSPICE

XSCHEM





Google + skywater TECHNOLOGY

Google + GLOBAL FOUNDRIES

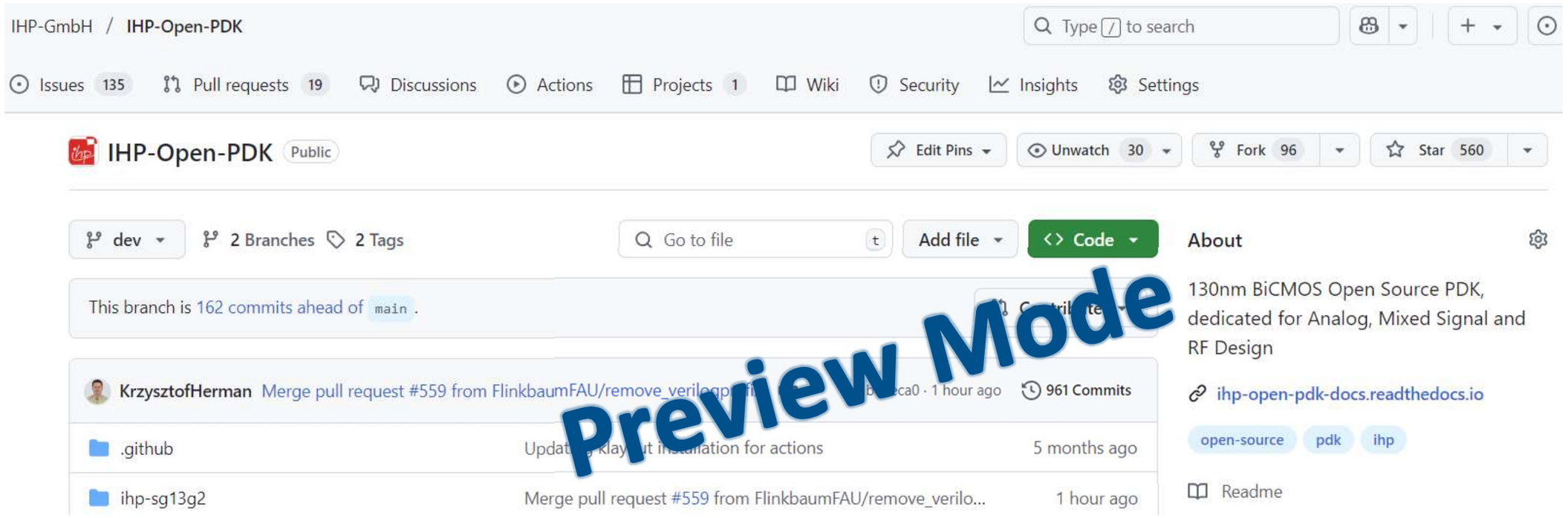
FOSS 180nm Production PDK
github.com/google/gf180mcu-pdk

IHP-Open-PDK

RISC-V®

test cases, regression tests, benchmarks, use cases,
user stories, feedback, error reports, feature requests

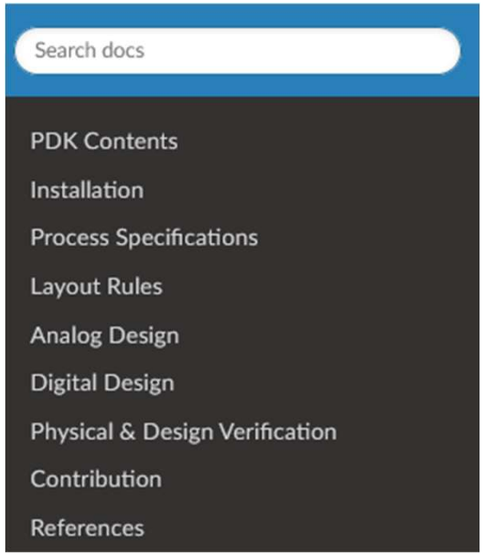
IHP Open PDK on github

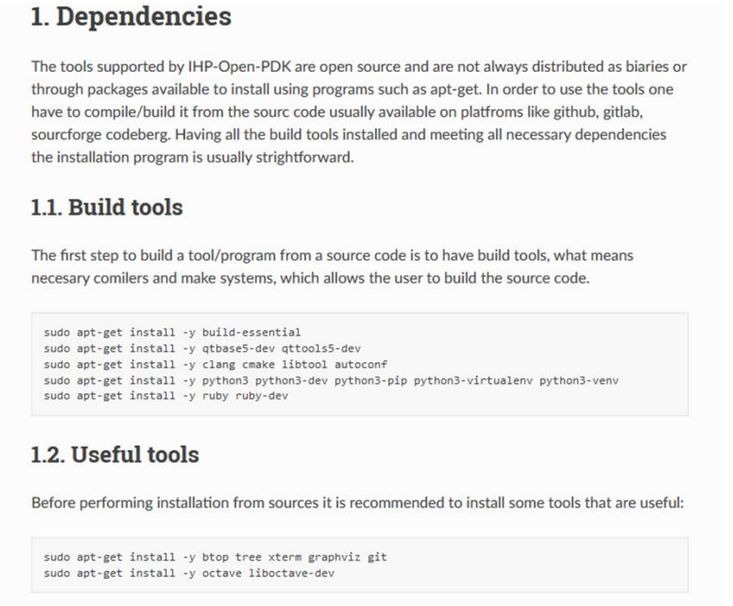


Minimal installation – plug and play approach

Working with the IHP Open PDK

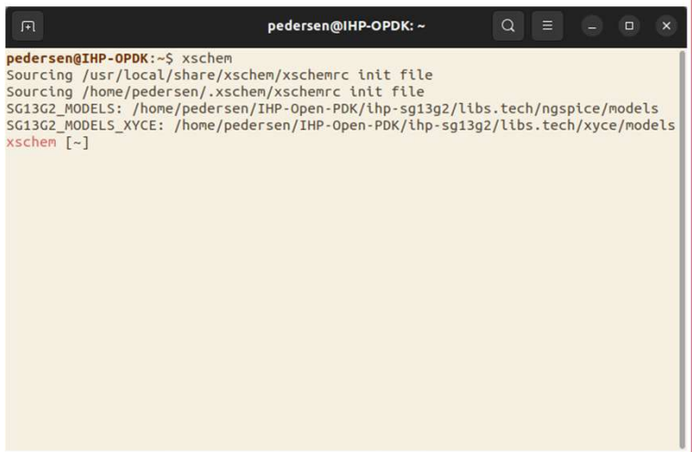


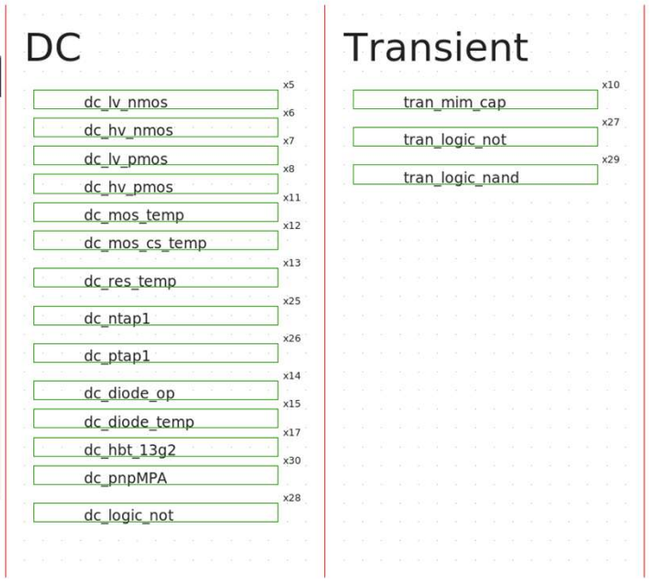


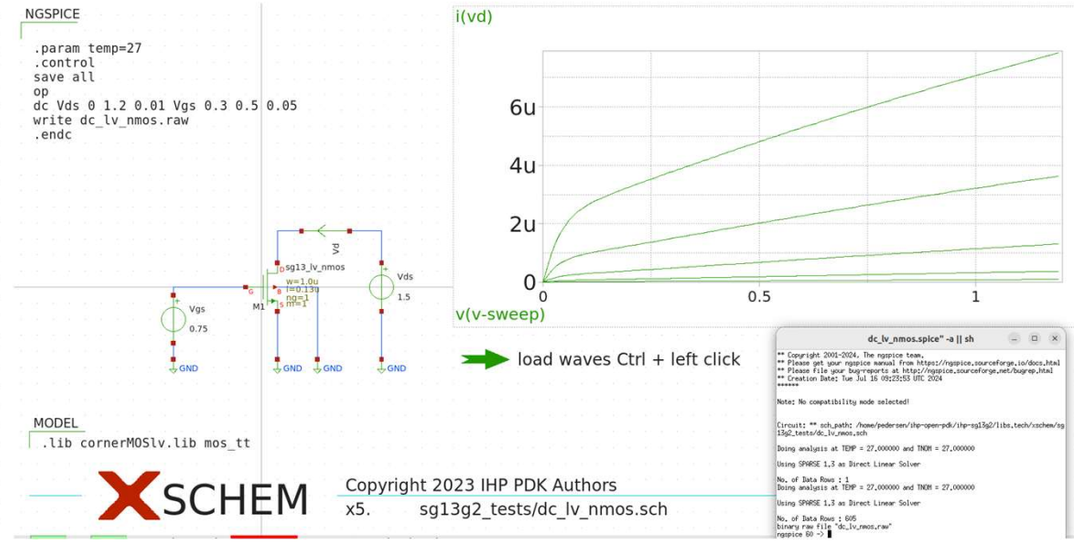


Ubuntu 22.04 + compatible
Configuration: $PDK_ROOT, $PDK
Dependencies installation: mainly apt-get
Tools have to be installed (binaries, sources)

Getting started really isn’t that hard!







Some of the Information Available in ChatGPT

1. Process Technology

- Overview of advanced semiconductor technology, including high-performance devices and integrated components
- Offers various modules for expanded functionality

2. Multi-Project Wafer (MPW) Services

- Includes details on pricing, schedules, and supported processes
- Specifies chip area requirements and approval process for smaller designs

3. Layout and Process Specifications

- Defines design rules, device specifications, and physical constraints
- Includes details on available materials and process layers



Design examples and courses

IHP-GmbH / IHP-AnalogAcademy

Code | Issues 32 | Pull requests | Actions | Projects

IHP-AnalogAcademy Private

main | 2 Branches | 0 Tags

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- Introduction to IHP Open PDK and SG13G2 Technology
- Foundations
- Bandgap Reference
- 50GHz Medium Power Amplifier
- 8-bit SAR ADC
- Final Thoughts

OS-EDA / Course

Code | Issues 3 | Pull requests | Projects | Wiki | Security | Insights

Course Public

main | 1 Branch | 10 Tags | Go to file

Course instance: 1 Week, hands on



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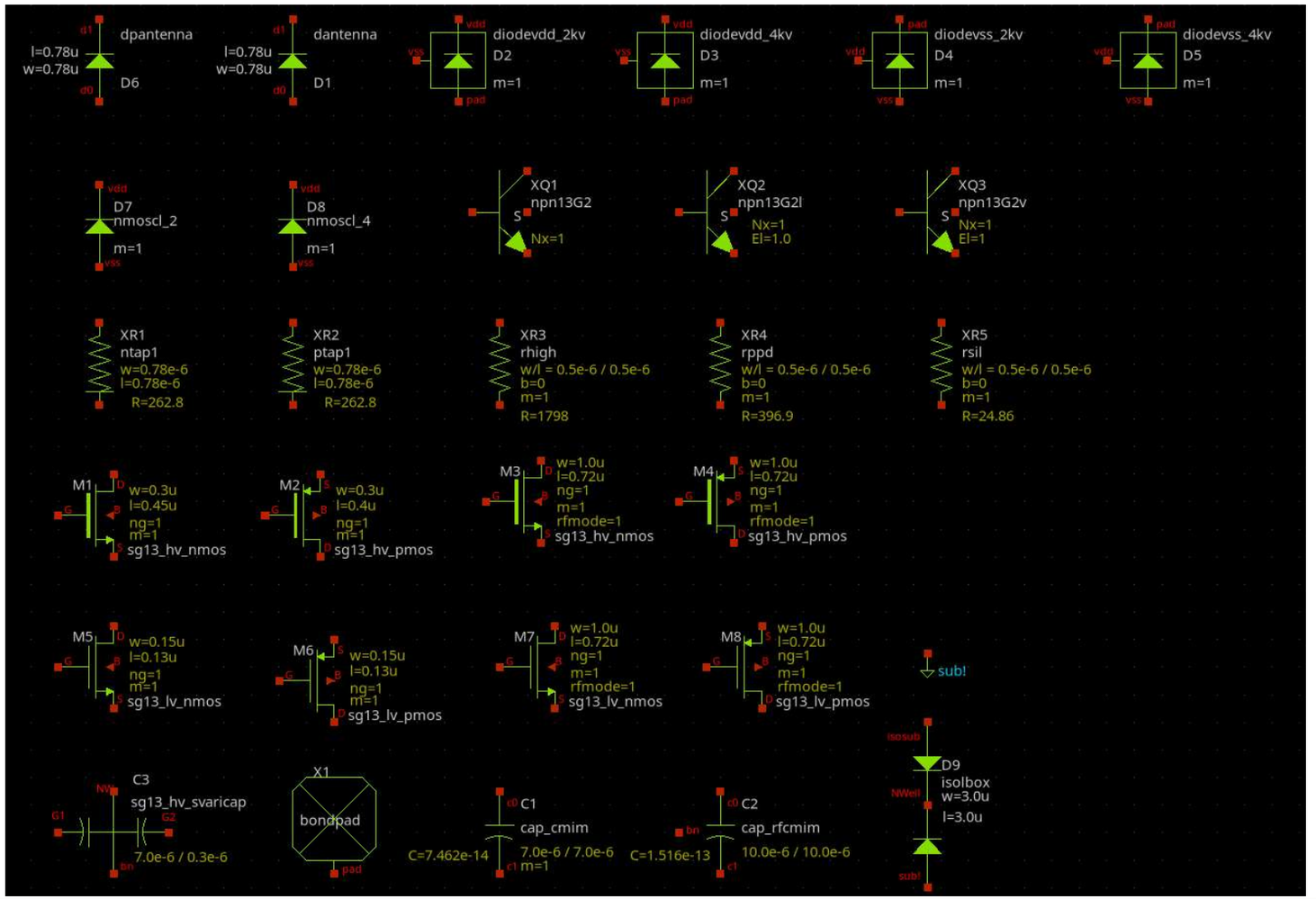
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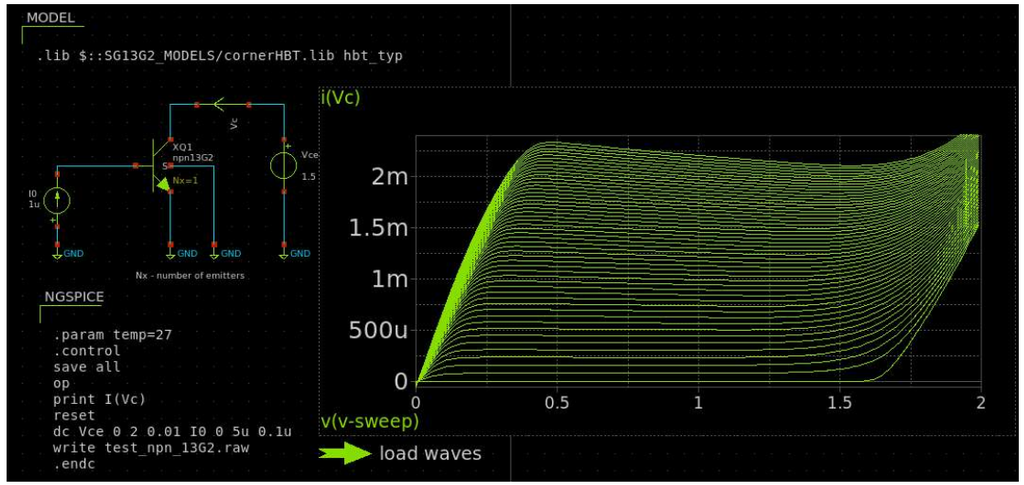
OpenPDK support for schematic capture

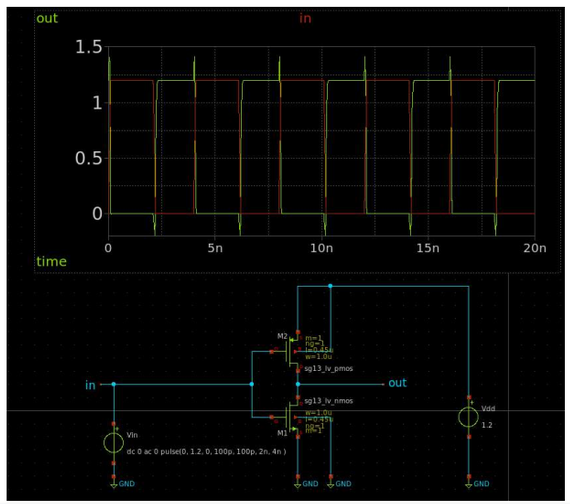


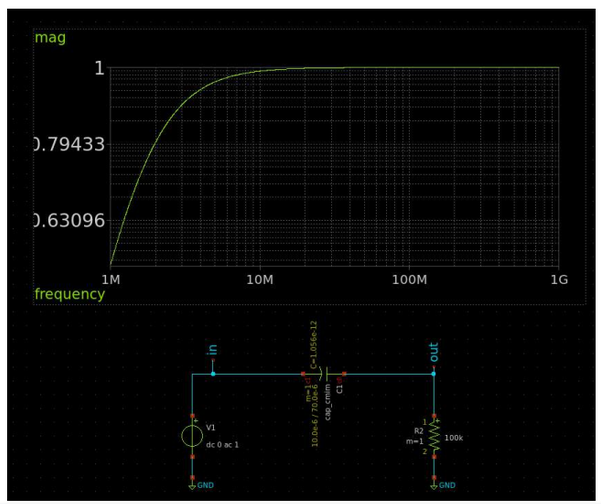
The current version of the IHP OpenPDK supports:

- xschem primitives for schematic capture
- automatic ngspice/Xyce compatible netlist generation
- example use cases to show the basic functionalities and parameters of the primitives

OpenPDK support for simulations



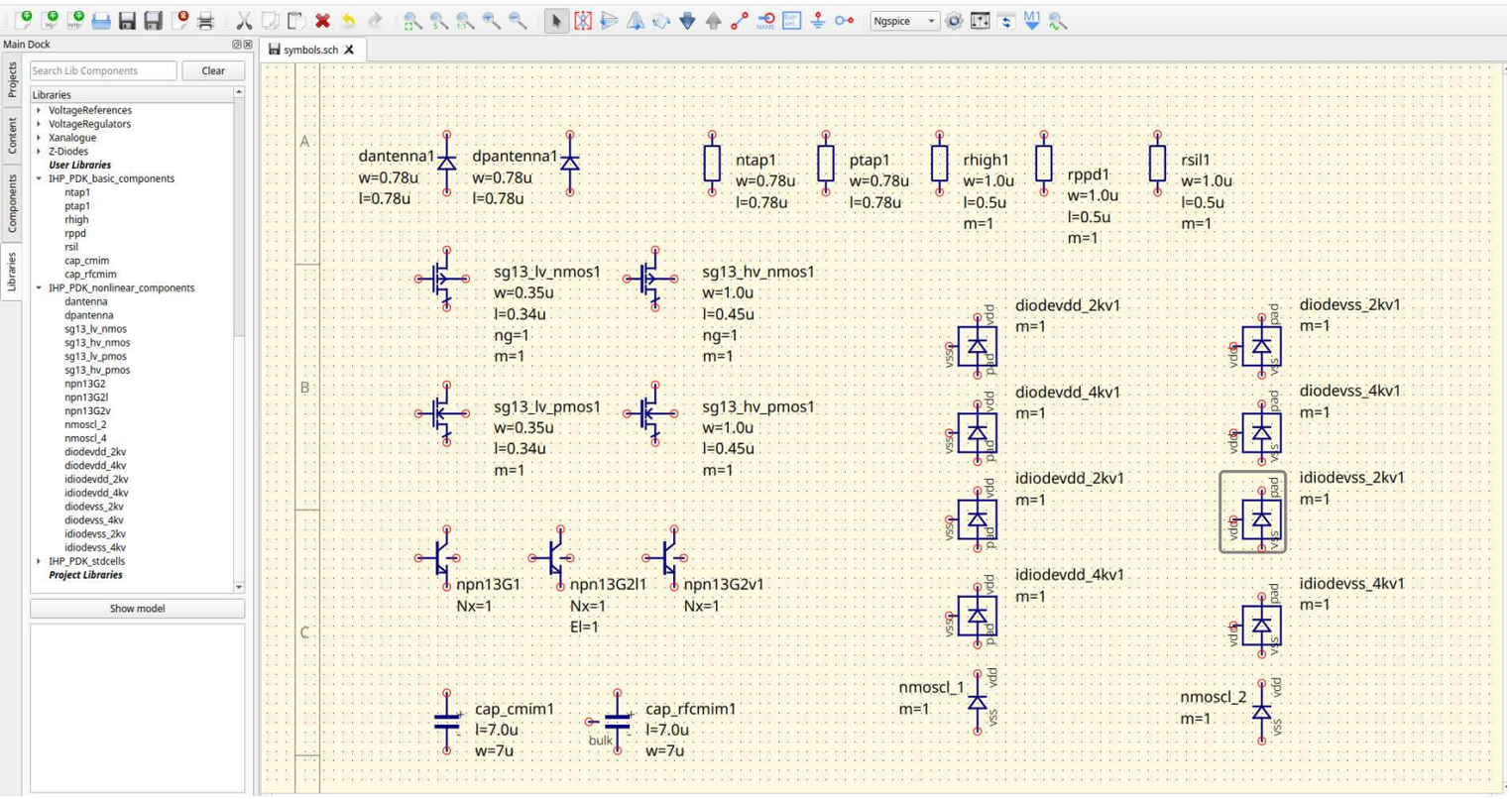




Analog simulation facilities:

- PSP103.6 MOSFET models from SemiMod,
- ngspice/Xyce compatible netlists,
- process corners: typical, best case, worst case,
- statistics related to the process variation,
- simulation examples segmented by simulation type: DC, TRAN, AC, MonteCarlo, S-parameters
- postprocessing python scripts,
- model extensions for RF frequency range,
- ngspice 40+ compatible

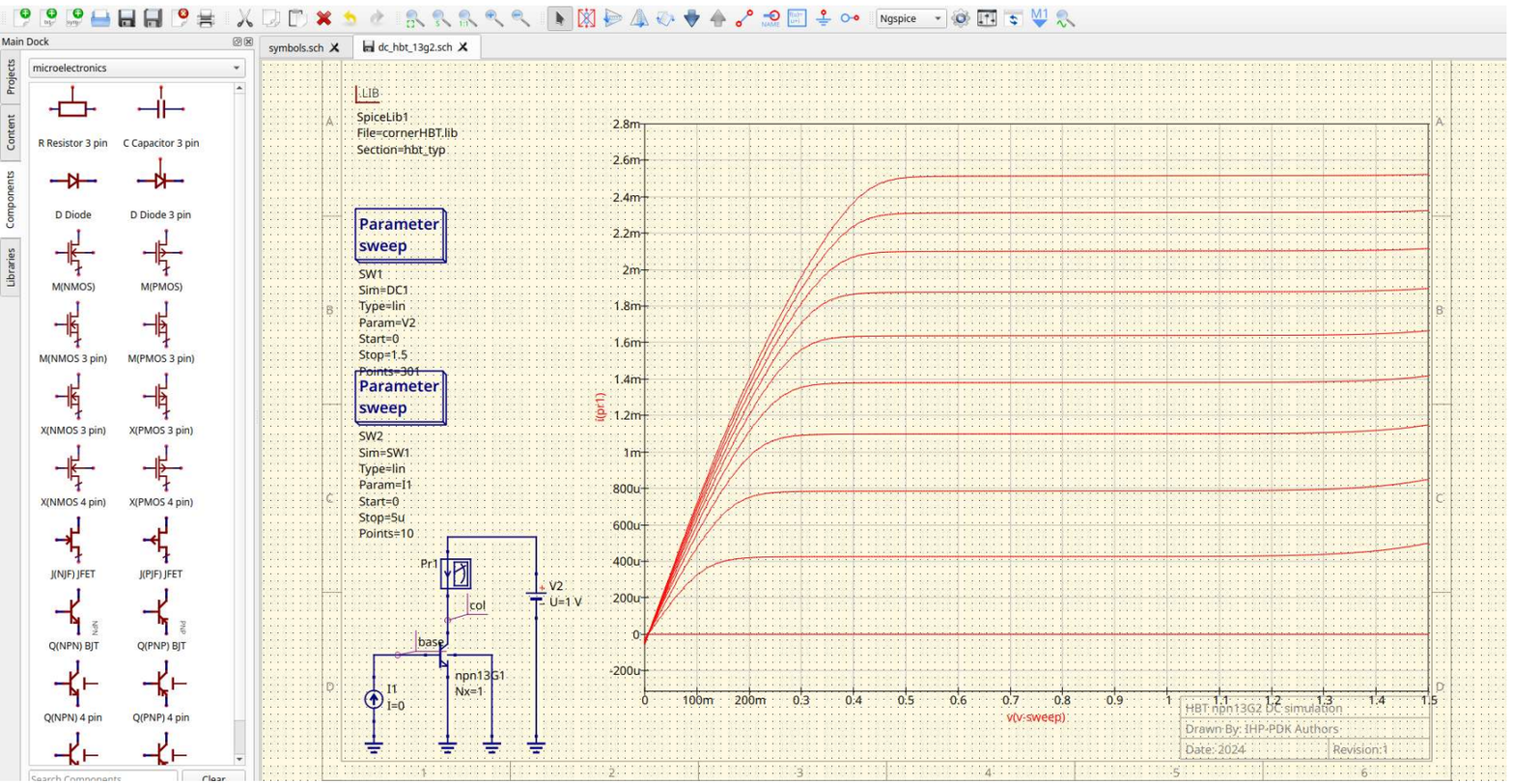
OpenPDK support for Qucs-S schematic capture



The current version of the IHP OpenPDK supports:

- primitives for schematic capture
- automatic ngspice/Xyce compatible netlist generation
- example use cases to show the basic functionalities and parameters of the primitives
- XSPICE model support (under development)

OpenPDK support for Qucs-S schematic capture

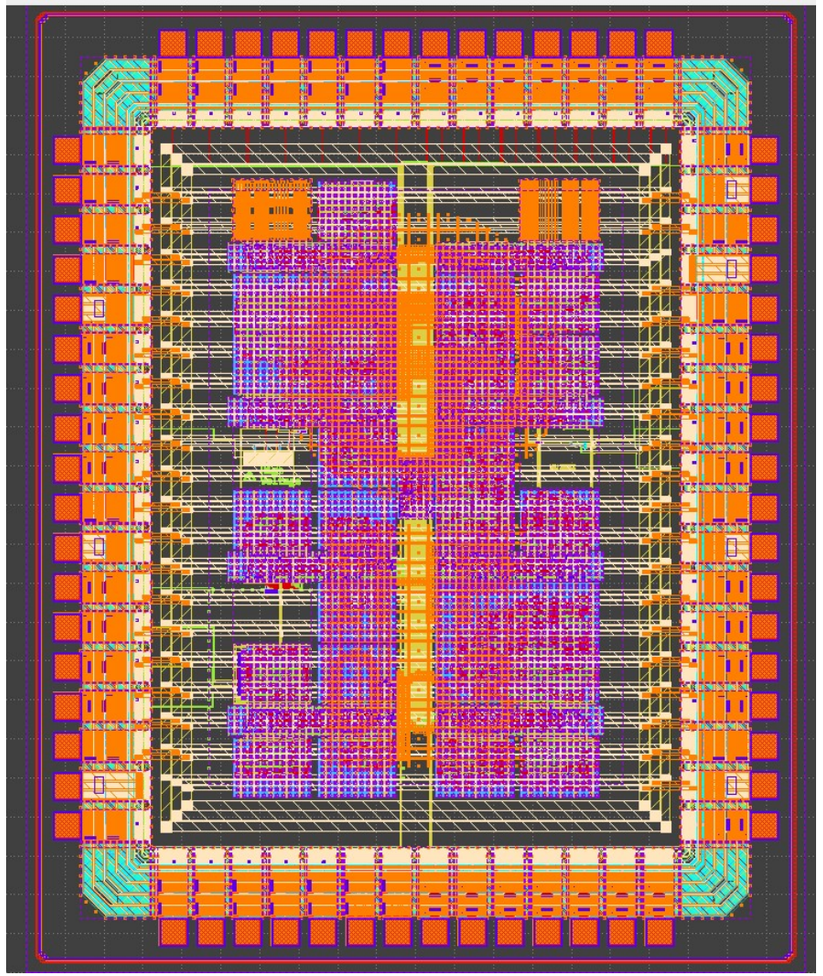


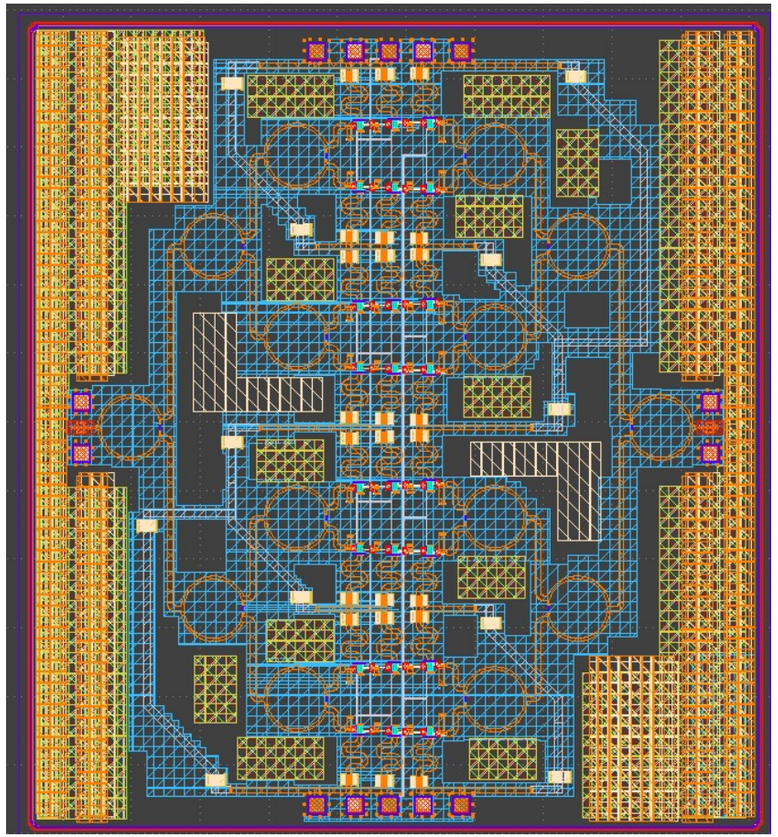
Qucs-S development:

- Agnostic support for PDK
- CDL netlisting
- Interoperability with OpenEMS and Klayout
- XML symbol library import
- RF related features

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KLayout – primary tool for open source layout design

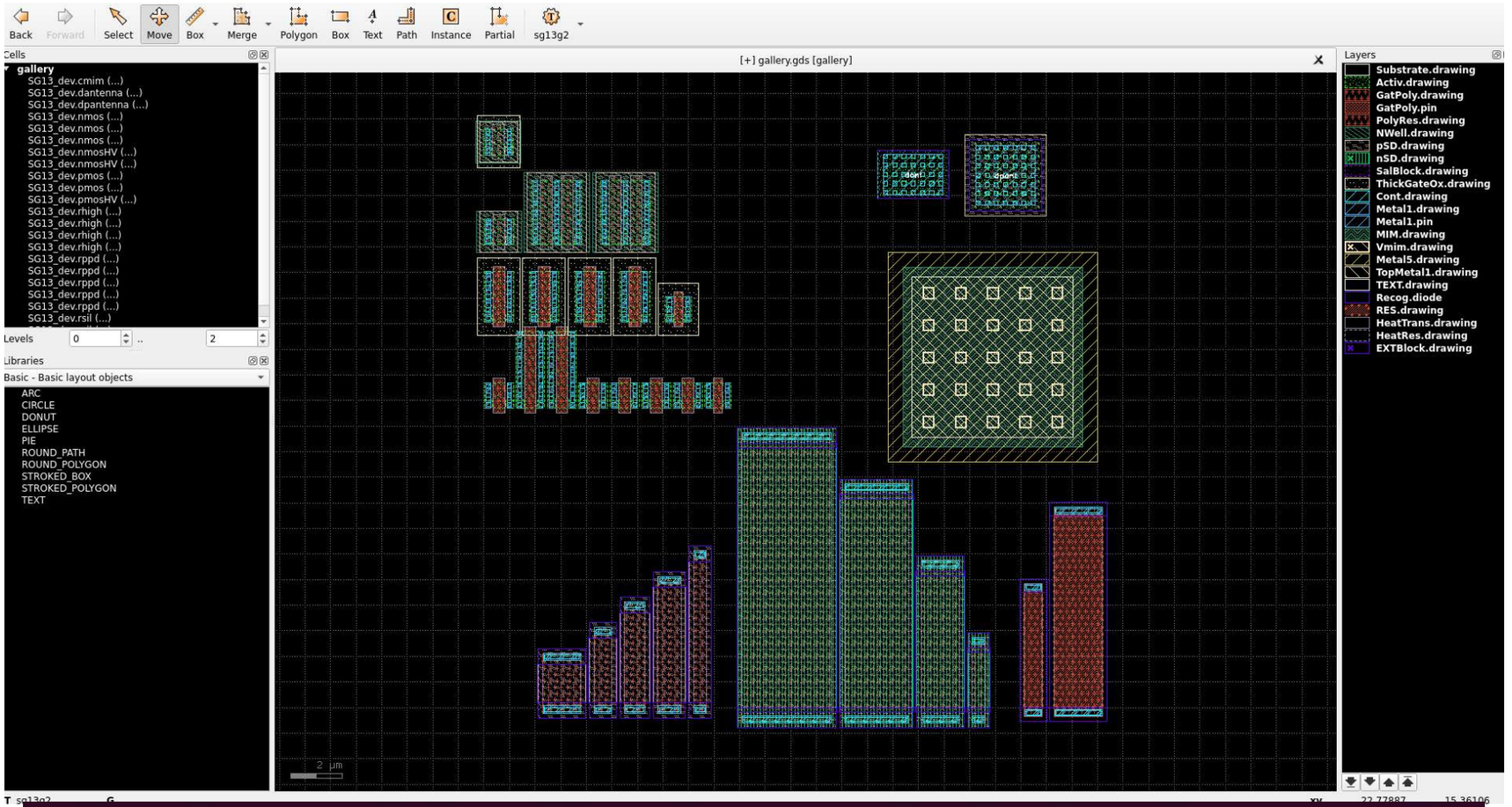




KLayout key features:

- hierarchical view
- parametric cell suport
- DRC/LVS checks
- command line batch mode
- XOR and DIFF tools
- Custom scripts in Python/Ruby
- Plugins

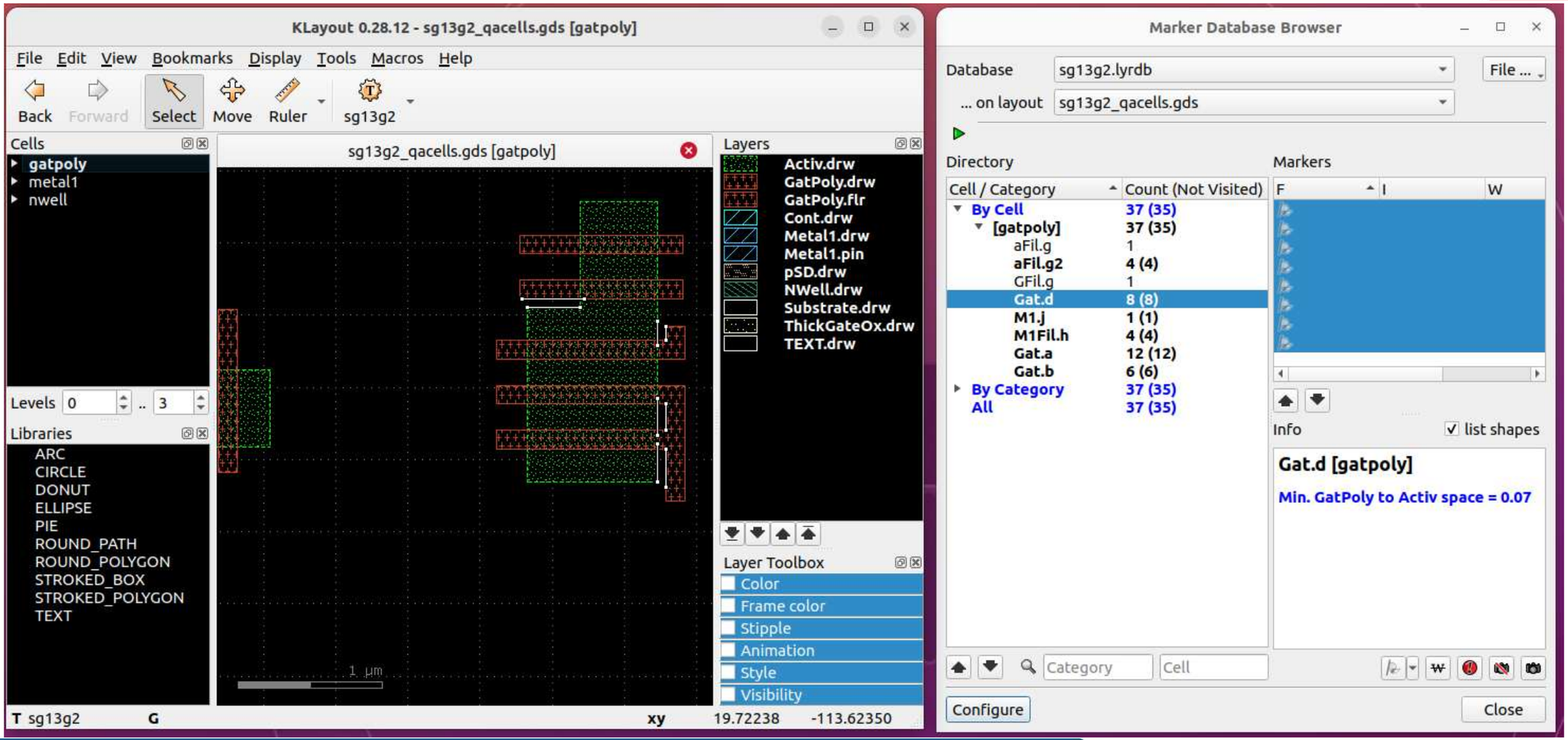
KLayout – Pycell support



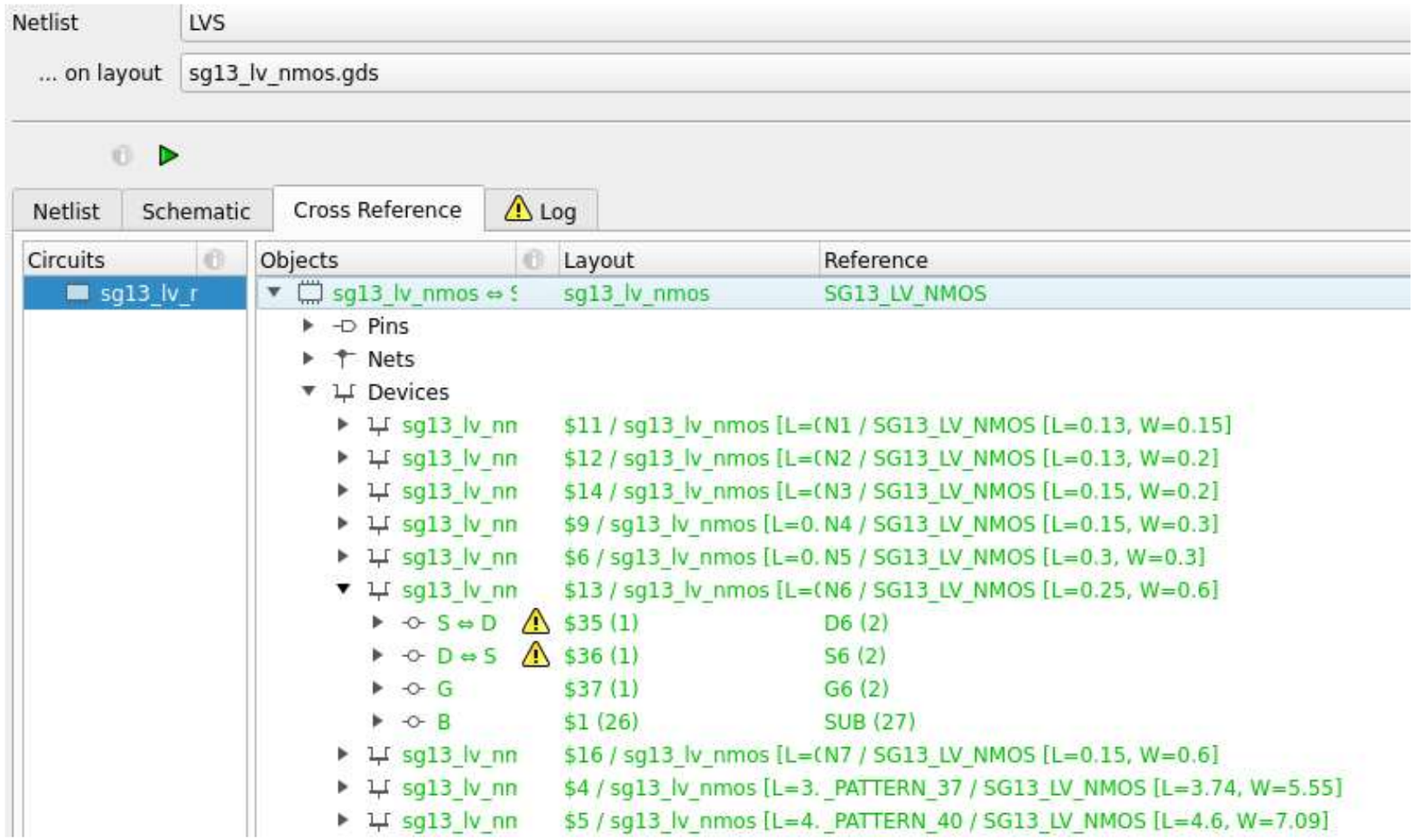

```
klayout -n sg13g2 -zz -r generator.py -rd netlist=netlist.spice -rd output=macros/gallery.gds
```

- Pycells compatible with Synopsys PyCell Studio and Keysight ADS
- KLayout wrapper API development
- Klayout Device generator

KLayout example DRC run on `gatpoly` QA cell



KLayout example LVS run on sg13_lv_nmos mosfets



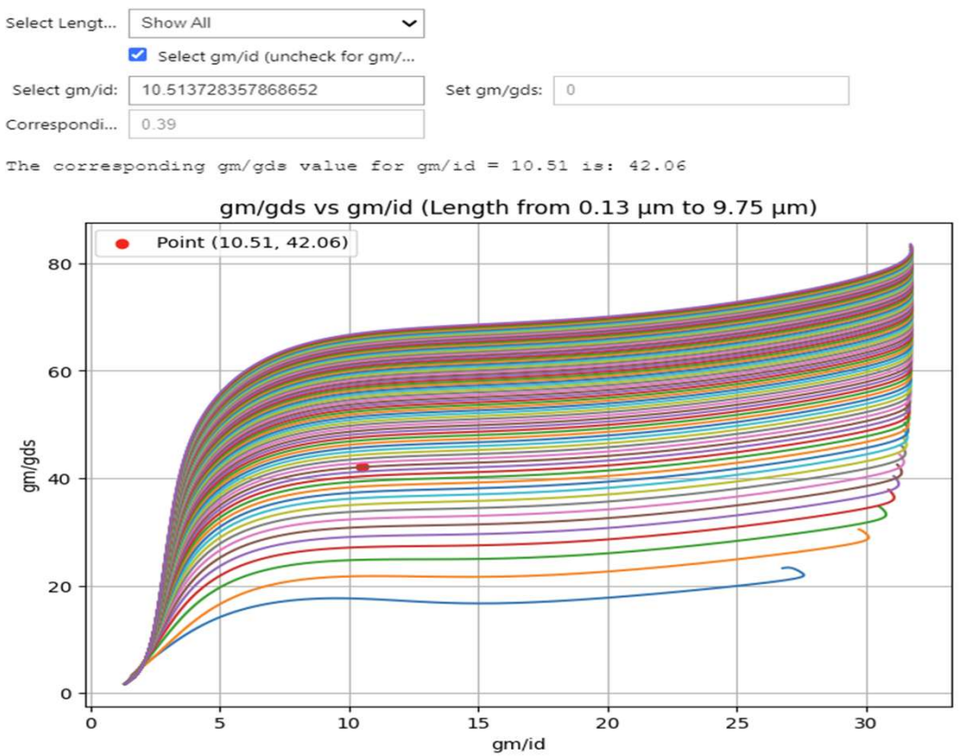
Klayout LVS ruledeck:

- accepts CDL netlist,
- extracts devices from layout,
- creates extracted netlist,
- performs checks and compares:
  - Pins
  - Nets
  - Devices
- reports inconsistencies
- can run in batch mode

OS tools for Advanced IC Design?

MOSFET Sizing with GM/ID Curves:

- Using Python scripting alongside Ngspice to generate GM/ID curves for efficient MOSFET sizing



Mixed Signal Design

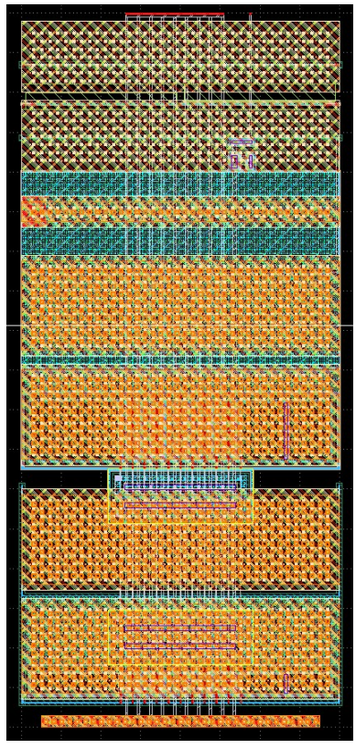
- Verilator
- Xspice
- Creating digital models in conjunction with analog design

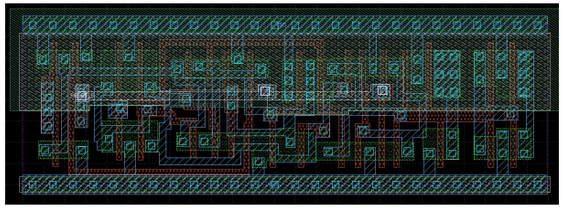
Layout Automation

- Physical verification, filler scripts
- Pcells generation from SPICE-Files
- Streamlining the layout process

Digital primitives

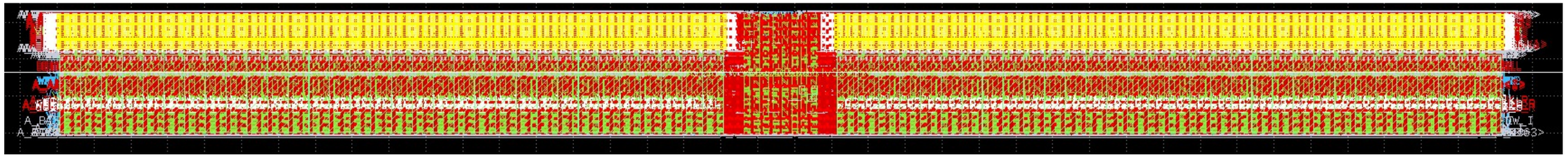
	StdCells	IOCells	SRAM
Verilog	x	x	x
LIB	x	x	x
LEF	x	x	x
CDL	x	x	x
SPICE	x	x	x
GDS	x	x	x



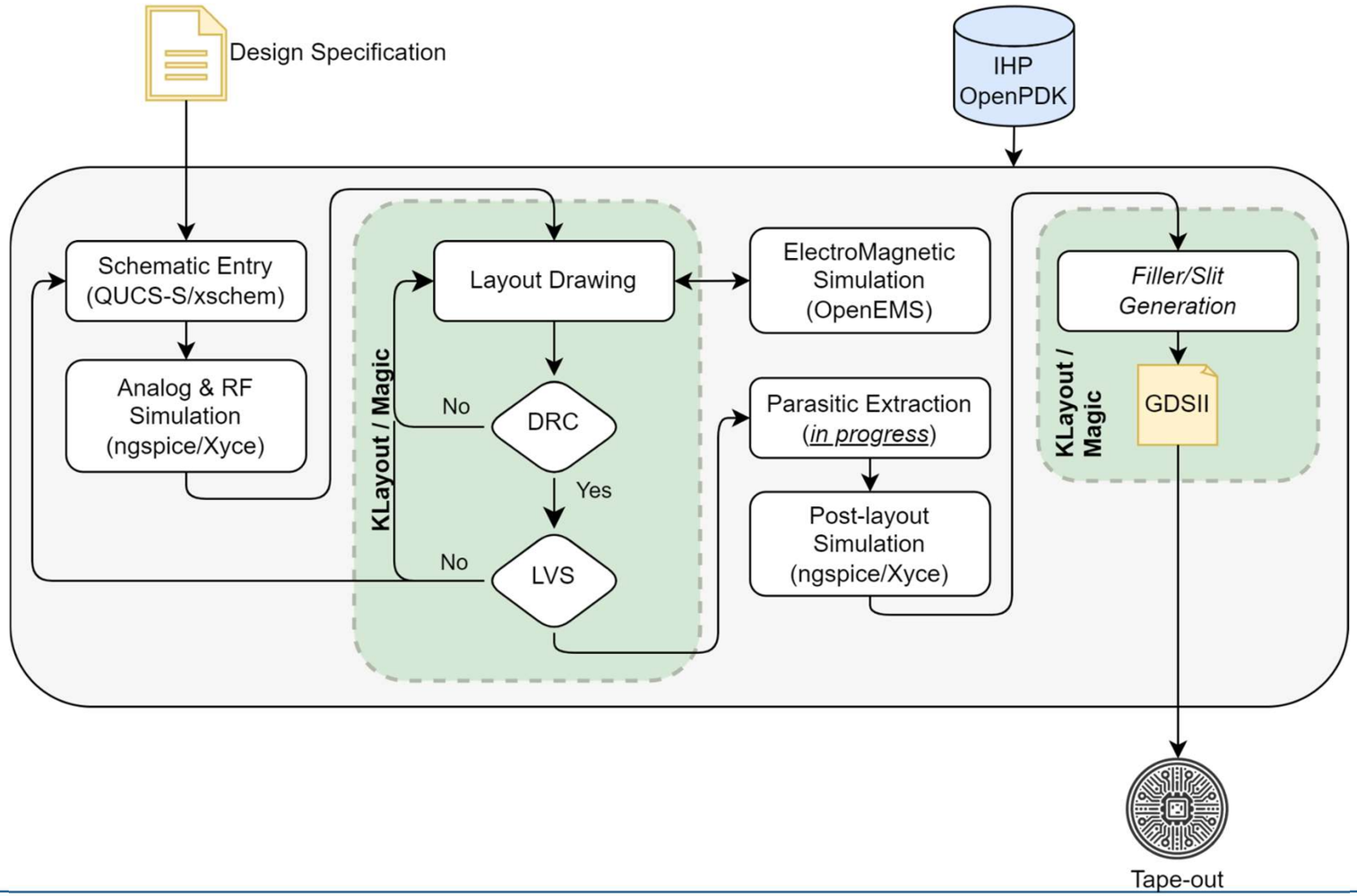


Digital components:

- stdcells
  - 84 cells,
  - combinatorial logic,
  - sequenntial elements,
  - scan flops,
  - gated cells.
- IOCells,
  - In, Out, InOut, Analog
  - different drive strengths
- SRAM
  - hard macros of a single port SRAM
  - different sizes

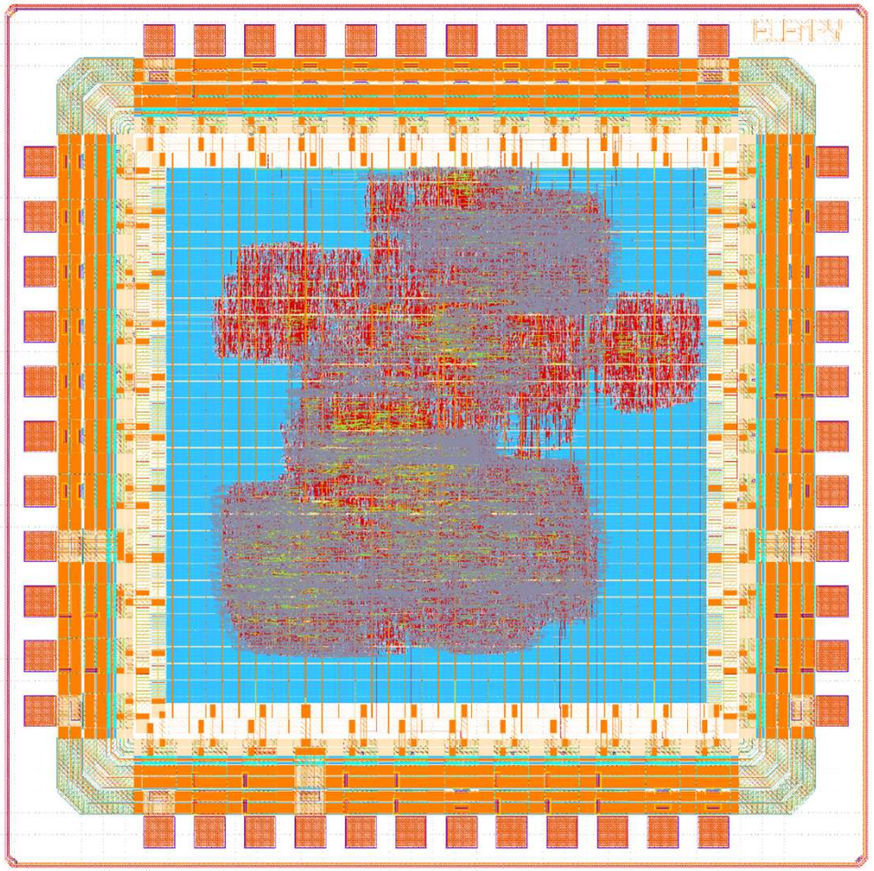


Analog/RF Open Source Design Flow: status at a glance

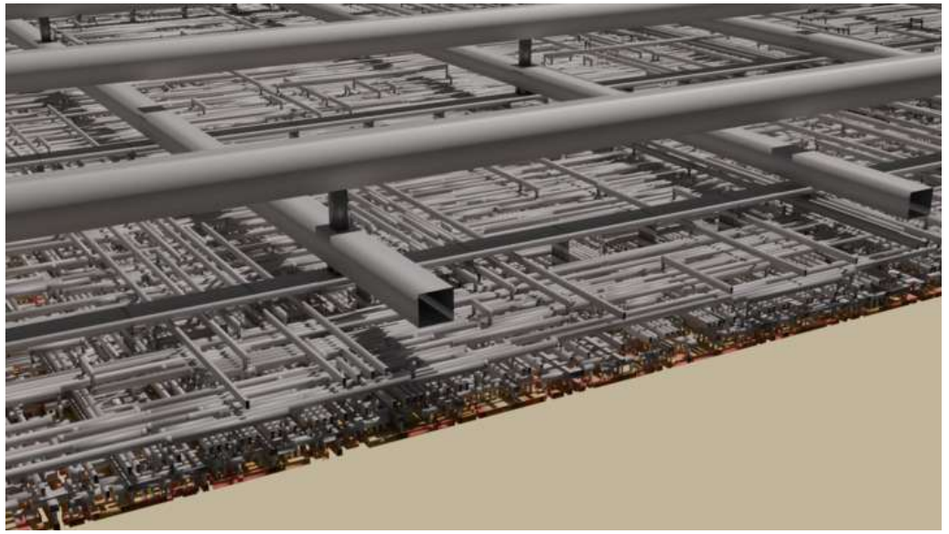


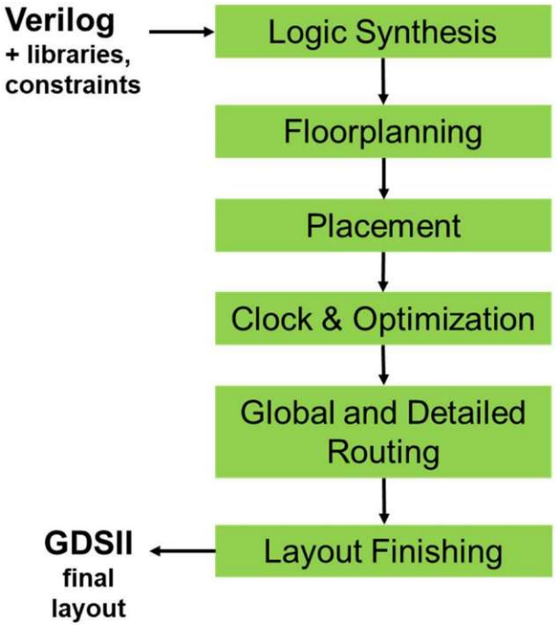
- KLayout-oriented flow
  - Layout design
  - Parameterizable cells
  - Physical Verification
- QUCS-S, xschem
- ngspice, Xyce
- OpenEMS, ElmerFEM, AWS Palace
- Working on a faster solver for EMS (ELMER)

Digital OpenROAD-flow-scripts Flow



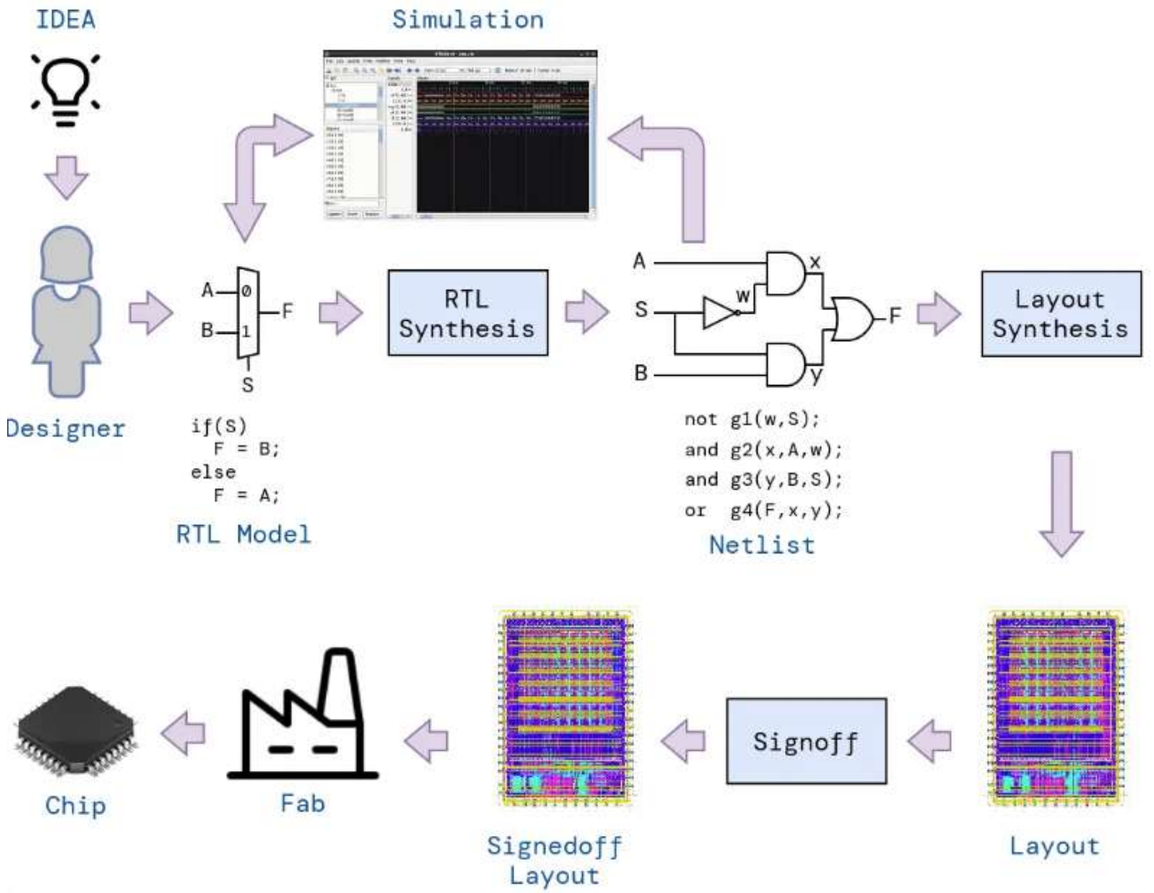
- Yosys + ABC - > synthesis
- OpenROAD -> PnR
- Klayout -> GDS streaming+checks

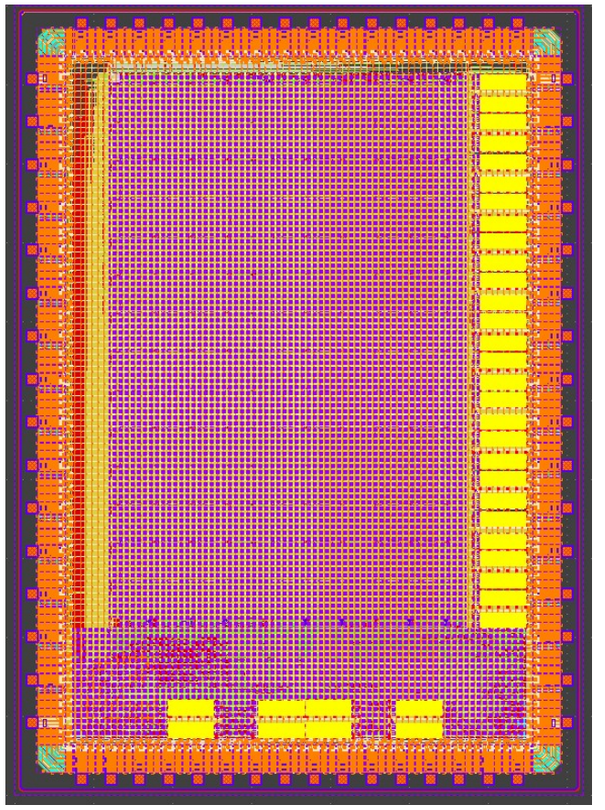


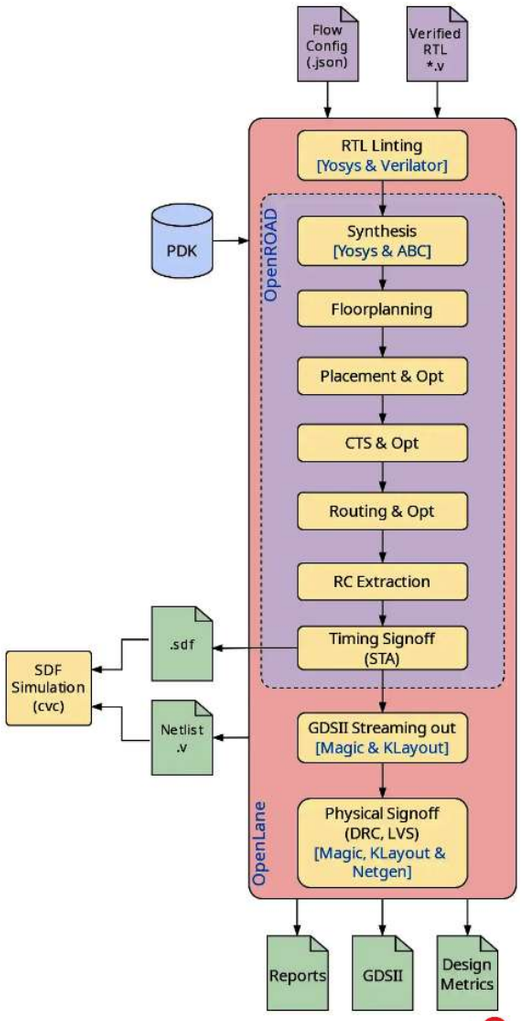


https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts

Digital LibreLane flow







https://github.com/librelane/librelane

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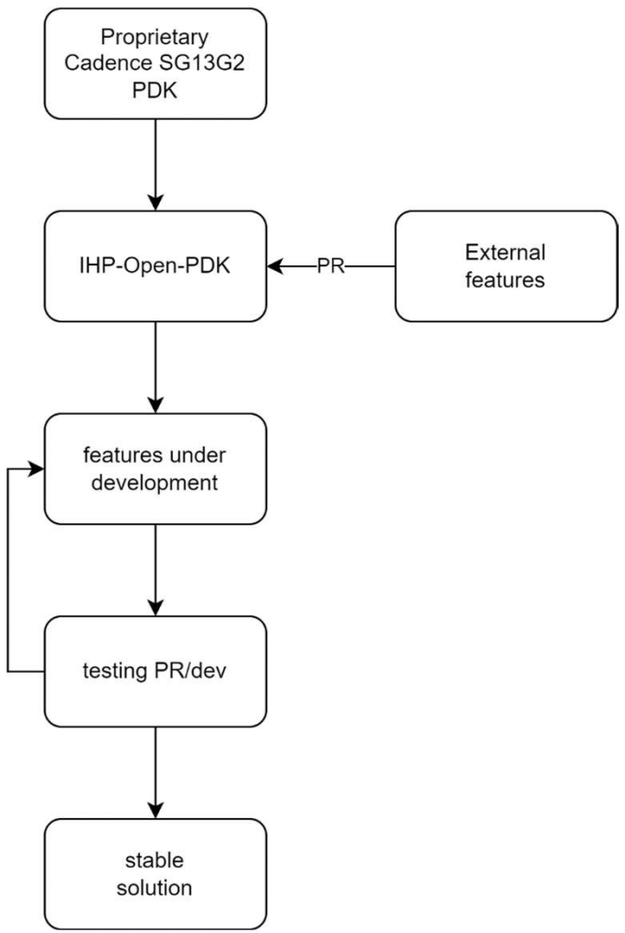
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PDK development cycle

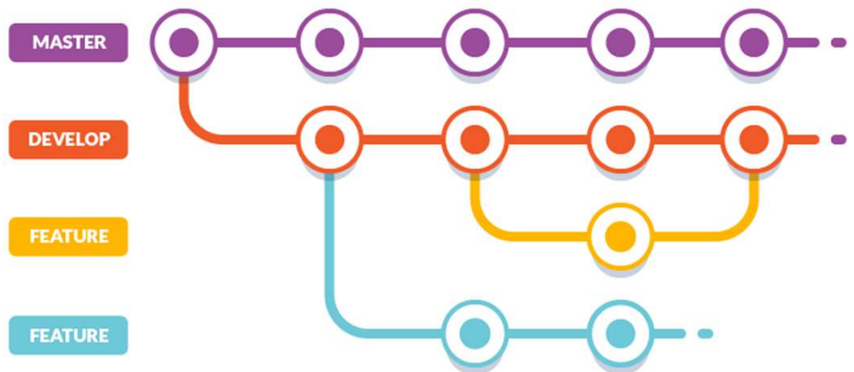


Principal rules

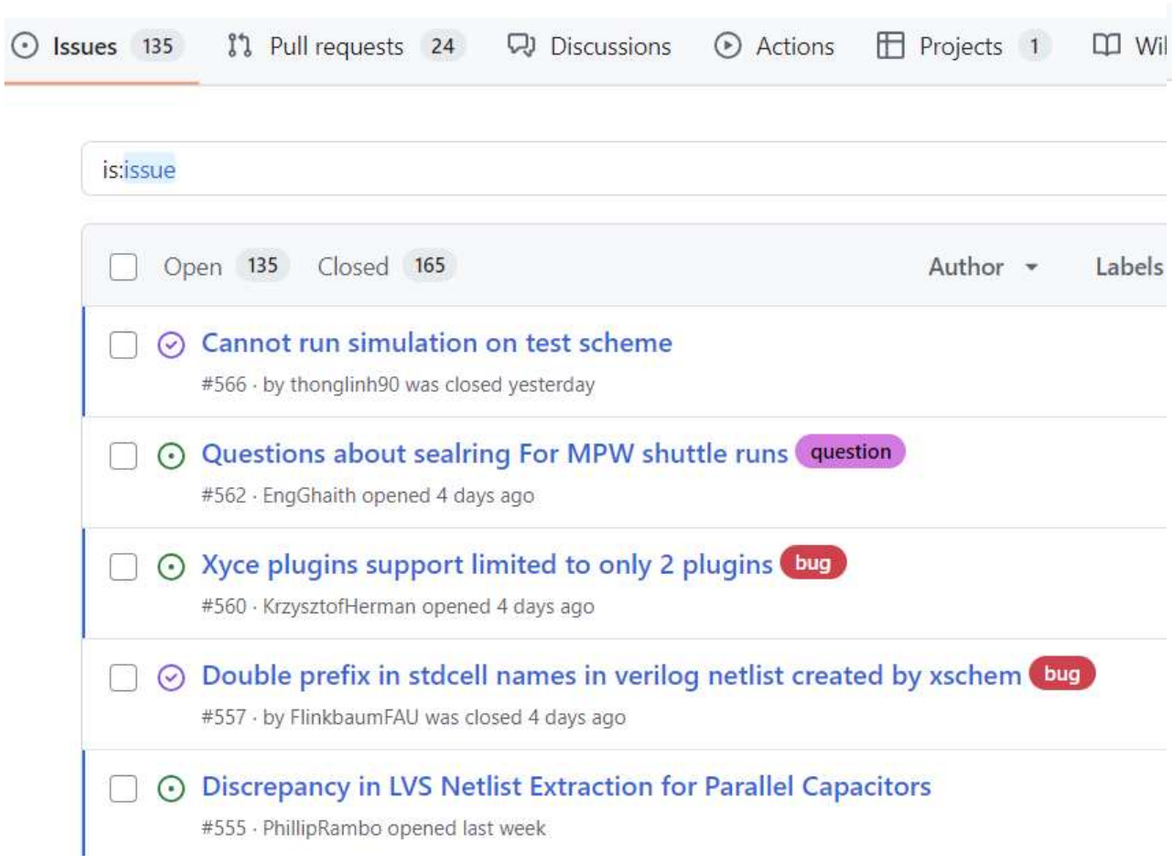
- Tight link between SG13G2 PDK and its open-source equivalent
- Compatibility with the open-source EDA tools
- Preserving the structure proposed in SKY130
- Read-only
- Based on git and following git flow

Key actors

- PDK core team
- Tools and flow developers
- Designers
- OS Community



PDK development cycle – community support



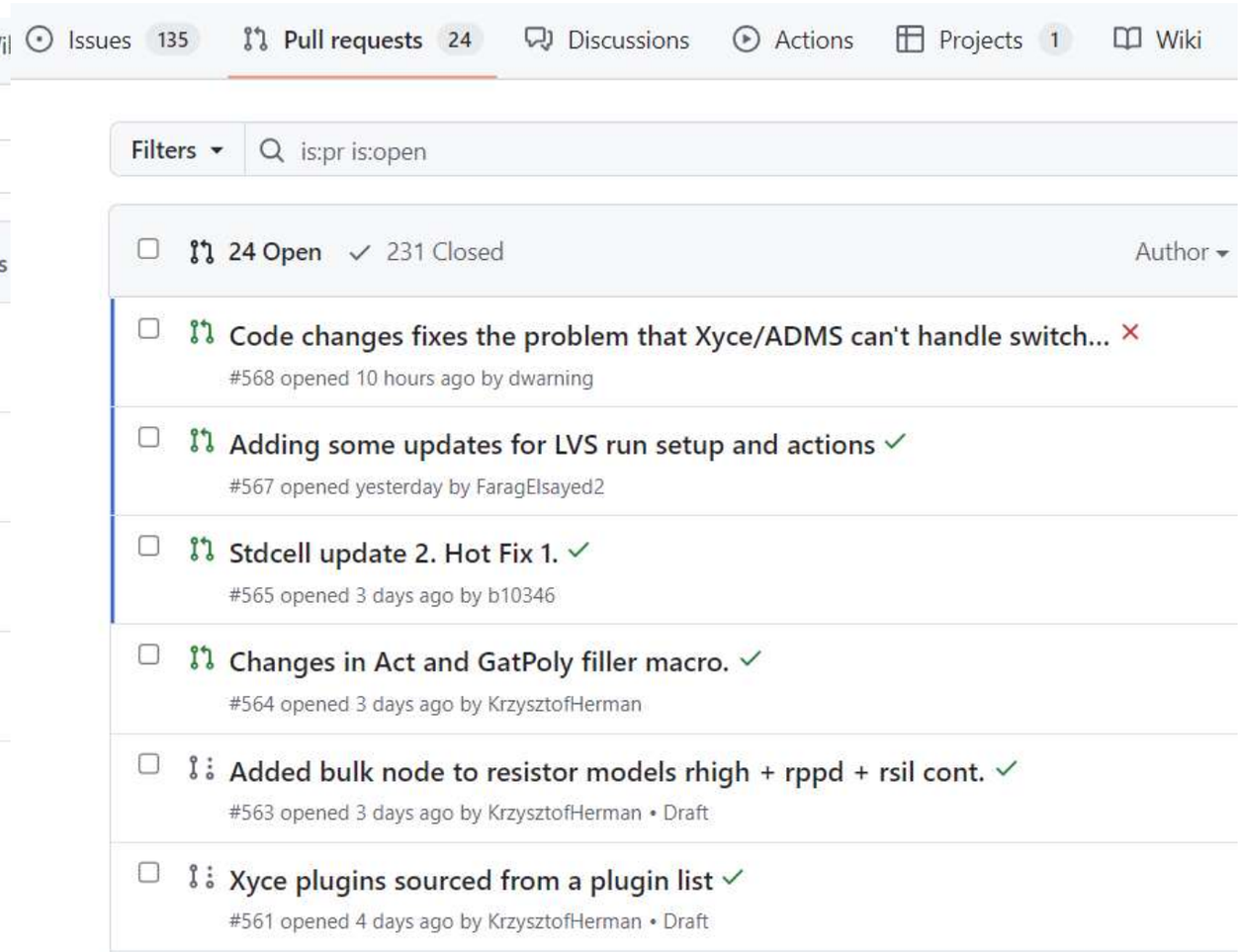


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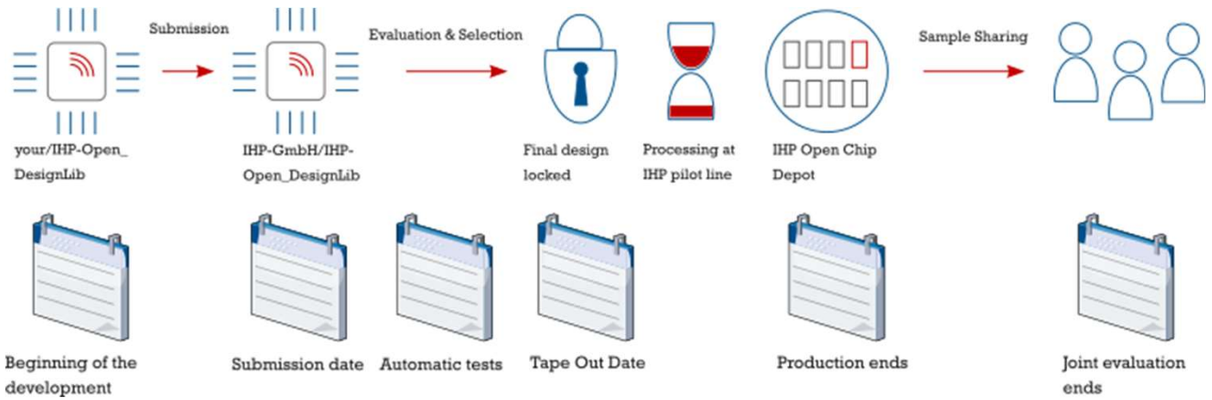
Leassons learned and future activities

Free MPW Runs - support open source PDK & design

- The table provides schedule of MPW Runs for FMD-QNC project in 2025 and 2025

Tape out date	22/05/24	11/11/24	22/11/24	01/03/25	09/05/25	18/07/25	15/09/25
Technology	SG13G2	SG13CMOS	SG13G2	SG13G2	SG13G2	SG13G2	SG13CMOS
Area [mm^2]	10	220	20	140	30	30	220

For more details check: https://ihp-open-ip.readthedocs.io/en/latest/



- Project funds can be used exclusively to produce chip designs for non-economic activities, such as university education, research projects, and others
- A concept for sustainable provision of free or low-cost MPW area for the open source community is to be developed.

Criteria for design IP selection from open community

Mandatory criteria for IP selection

- Completeness of IP data (all data need to be open source)
- DRC error free designs
- Area below 2 mm2 preferred (larger designs only if area is available)
- Potential export restrictions

Additional criteria for IP selection

- First time submission (preferred)
- Design should use open source tools supported by IHP open PDK
- For SG13G2 runs designs using SiGe (preferred)
- Documentation quality
- Uniqueness, not yet seen designs (i.e. if there were no ADCs before, an ADC design would get a higher point)

Flow to upload designs for MPW run

- Before a design can be considered for fabrication start a pull request at https://github.com/IHP-GmbH/TO_<date>
- All design data need to be submitted at **least 2 weeks** before TAPE OUT
- Europractice will check designs regarding mandatory criteria (Completeness, DRC clean GDS, …)
- Europractice registers area and upload selected GDS files
- Designs will be processed by IHP
- IHP can rent samples for joint evaluation **under certain agreement (Continuation on next slide)**
- Evaluation results need to be published on https://github.com/IHP-GmbH/IHP-Open-DesignLib
- Evaluation to provide permanent samples are under evaluation

The principal goal is to build an open source design library for future projects

Low Cost Open MPW Runs

In order to maintain open source MPW IHP plans to have two OpenMPW runs per year

- SG13CMOS – CMOS only run with AL BEOL where price „P” in EUR is calculated based on total area (mm^2) „A” booked

$$P = \begin{cases} 1500 & if \quad A < 100 \\ 150000/A & if \quad 100 \leq A \leq 150 \\ 1000 & if \quad A > 150 \end{cases}$$

- SG13G2 – fully featured G2 with AL BEOL at approx. 2800 EUR/mm^2

The above mentioned prices refer to the open-source designs, where all the views are compatible with the open source EDA tools.

For customers who do not wish to disclose their IP, we offer participation in the OpenMPW program at a 20% discount off the regular price, as the wafers can be shared with other customers.

The turn around processing time is approx. 6 months for CMOS - 8 months for BiCMOS

The typical offer includes 20 bare die samples. Packaging will be offered on request (additional fee can be applied)

The first run can take place Nov 2025 (CMOS)

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Key takeaways

- Strong open-source community and interest
- The open source enables and facilitates education and collaboration
- OS tools develops your understanding of semiconductor manufacturing
- Chip design demystification - hands on experiences
- EDA tools well oriented to support open source PDK’s, but still a lot to be implemented (get involved)
- Open source Design library is still missing (get involved!)
- Our team is growing, check out the open positions

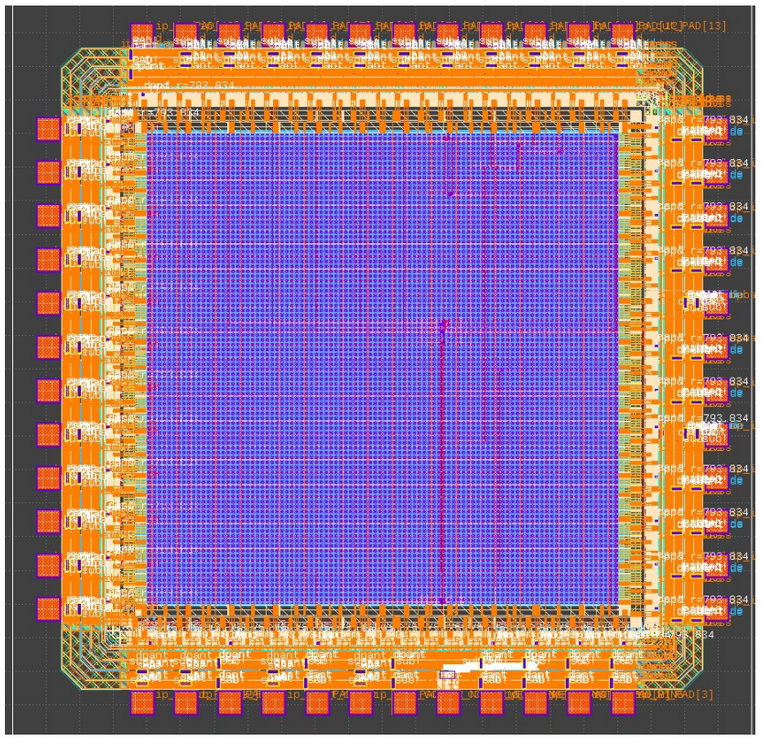
Work in progress – analog/mixed/RF flow

- Parasitics Extraction PEX – ongoing, https://github.com/martinjankoehler/klayout-pex
- Noise modeling in ngspice (transient noise, low frequency noise) – ongoing
- Qucs-S support – ongoing, agnostic PDK support, RF features
- Device models – issues found and reported by community members, migration to Verilog-A behavioral models
- New and fast DRC ruledeck for klayout
- OpenEMS integration for EM filed solving
- Mixed mode testcases using xspice + verilator

Full chip design example

- OpenROAD-flow-scripts does not provide full chip design example
- Padring design can be a tedious thing
- https://github.com/KrzysztofHerman/orfs-design-template

Cell / Category	Count (Not Visited) - Waived
By Cell	102 (101)
[ihp_top]	102 (101)
AFil.g	1 (1)
AFil.g2	11 (11)
M1.j	1 (1)
M2.j	1 (1)
M3.j	1 (1)
M4.j	1 (1)
M5.j	1 (1)
M1Fil.h	11 (11)
M2Fil.h	36 (36)
M3Fil.h	14 (14)
M4Fil.h	14 (14)
M5Fil.h	9 (9)
TM2.c	1
By Category	102 (101)
All	102 (101)

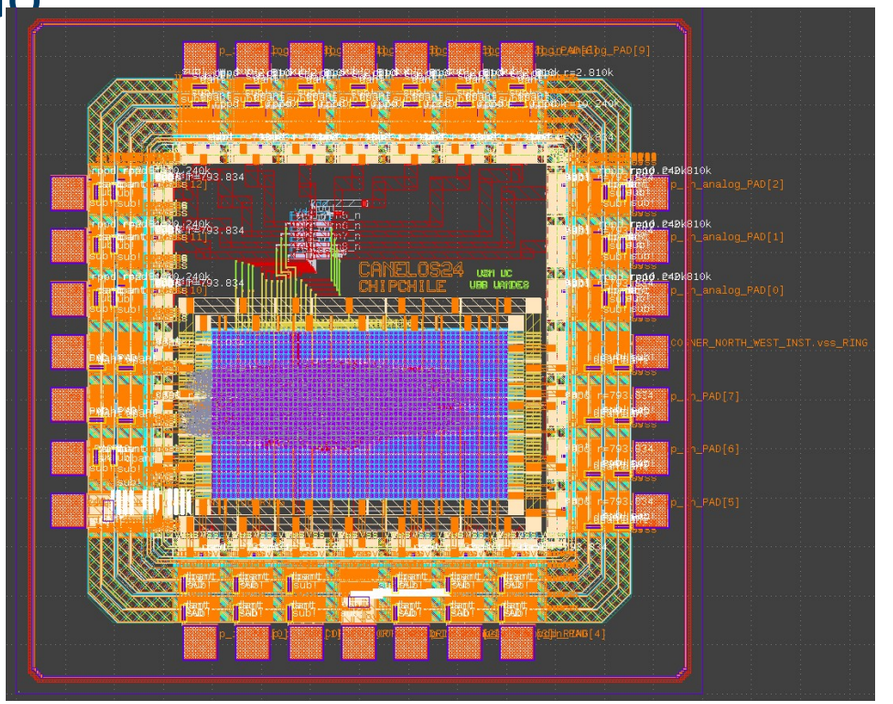


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Mixed signal design using Ngspice

- NGSPICE is capable to handle a mixed mode simulation using shared object generated by Verilator from Verilog bahavioral code.
- https://canelosworkshopihp2025-docs.readthedocs.io

- 8:1 analog mux
- SPI interface
- neural network



How to reach us

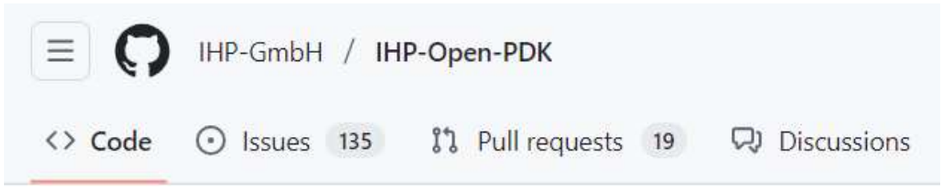
Email

- openpdk@ihp-microelectronics.com



GitHub

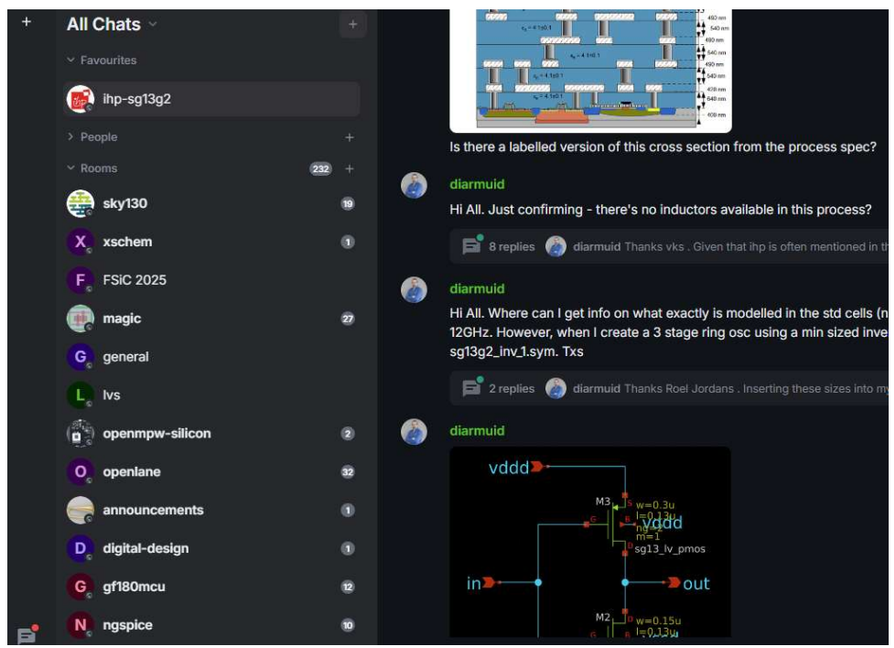
- issues – for reporting issues, questions
- discussions – for requesting features



Open Source Fossi-Chat.org channel:

`#ihp-sg13g2`

- general discussions
- announcements

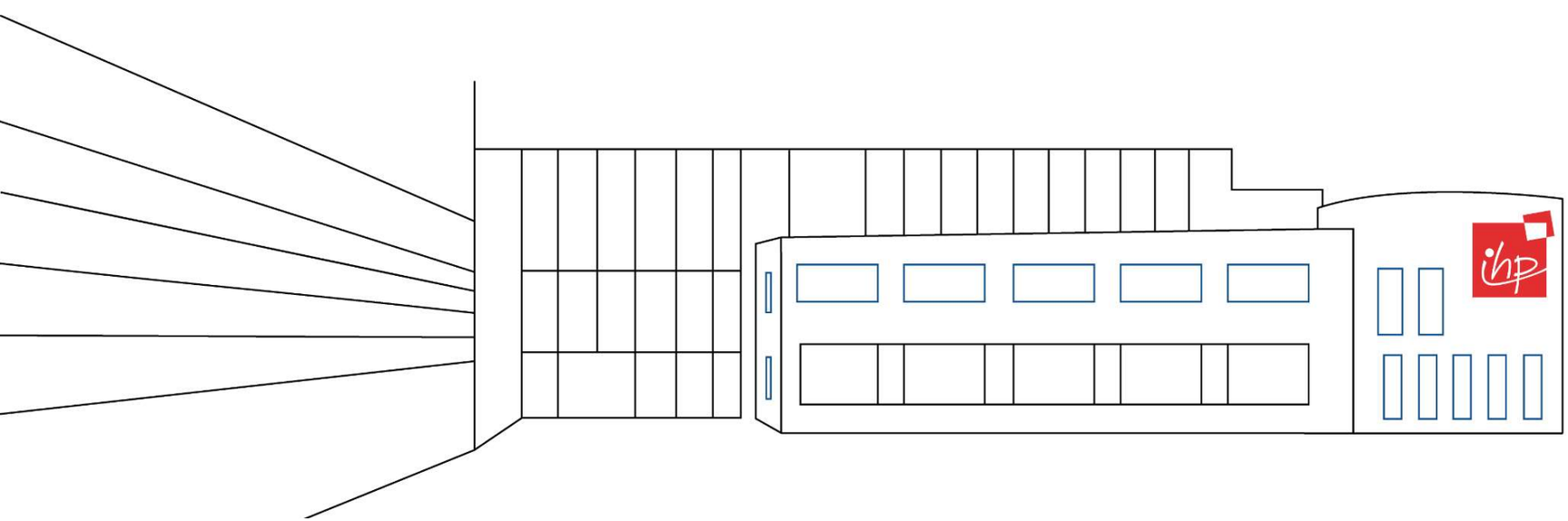


Acknowledgment

- Thanks to my colleagues at IHP and all the people involved in the development of the PDK
- And special thanks to the following public founded German projects:
  - VE-HEP (16KIS1339K) https://elektronikforschung.de/projekte/ve-hep-1
  - IHP Open130-G2 (16ME0852) https://www.elektronikforschung.de/projekte/ihp-open130-g2
  - FMD-QNC (16ME0831) https://www.elektronikforschung.de/projekte/fmd-qnc

Thank you for your attention!

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